S. J. P. N. TRUST'S

HIRASUGAR INSTITUTE OF TECHNOLOGY, NIDASOSHI

Accredited at 'A+' Grade by NAAC

Programmes Accredited by NBA: CSE & ECE

Department of Computer Science & Engineering

MACHINE LEARNING

Module 2:

Chapter 2: Understanding of Data-2

Dr. S. V. Manjaragi

Asso. Prof. , Dept. of Computer Science & Engg.,

Hirasugar Institute of Technology, Nidasoshi

Bivariate Data

- Bivariate data contains **two variables.**
- Bivariate data deals with causes of **relationships.**
- The aim of bivariate analysis is to **find relationships among variables.**
- The relationships can then be used in **comparisons, finding causes,** and in further explorations
- Eg: Dataset

Table 2.3: Temperature in a Shop and Sales Data

Temperature (in centigrade)	Sales of Sweaters (in thousands)
5	200
10	150
15	140
20	75
22	60
23	55
25	20

Bivariate Data Visualization

- **Scatter plot** is used to visualize bivariate data.
- It is useful to plot two variables with or without nominal variables, to **illustrate the trends, and also to show differences.**
- It is a 2D graph showing the relationship between two variables.
- **The scatter plot indicates strength, shape, direction and the presence of Outliers.**

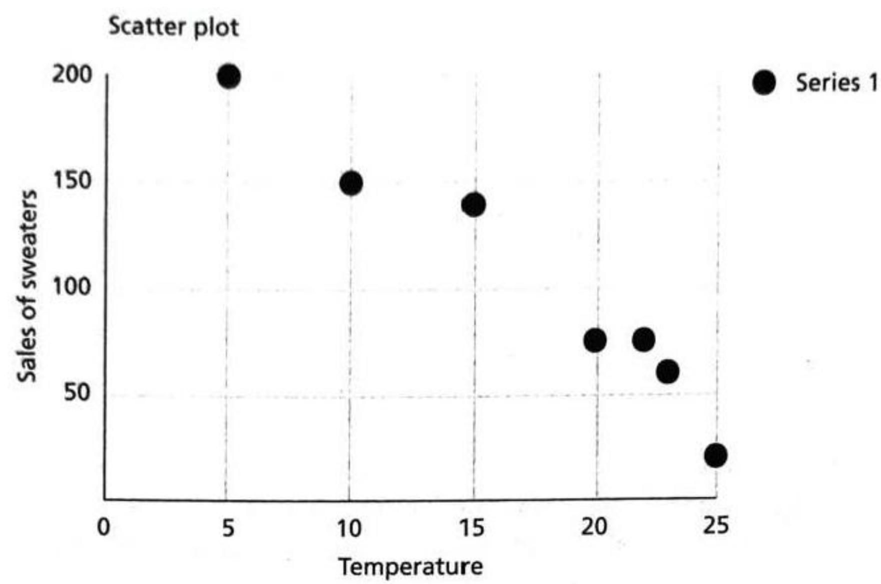


Figure 2.11: Scatter Plot

Bivariate Data Visualization

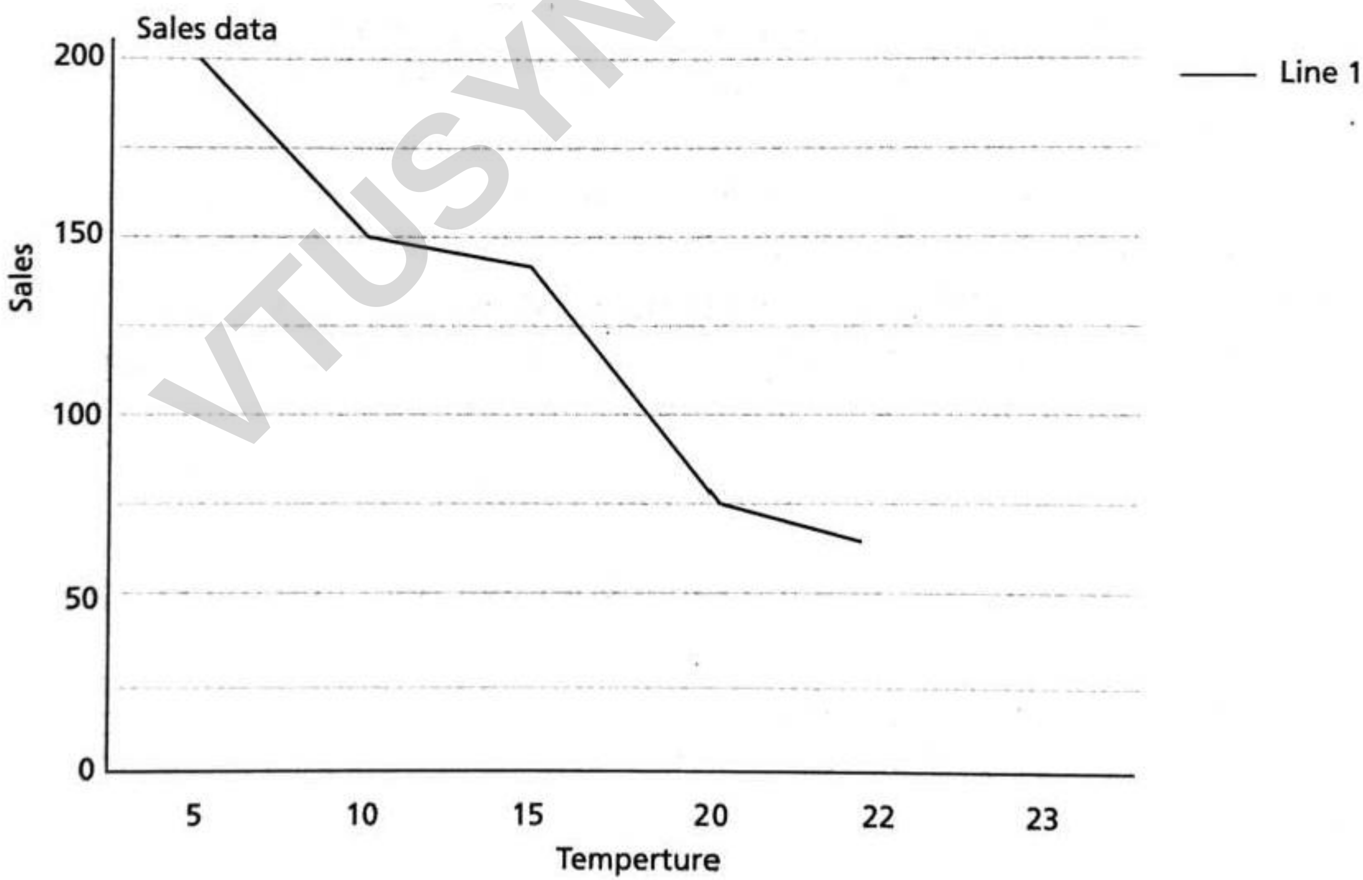


Figure 2.12: Line Chart

Bivariate Statistics

- **Covariance and Correlation** are examples of bivariate statistics.
- **Covariance** is a measure of **joint probability of random variables, say X and Y.**
- It is defined as **covariance(X, Y) or COV(X, Y)** and is **used to measure the variance between two dimensions.**
- The formula for finding co-variance for specific *x,* and y are:

$$\operatorname{cov}(X, Y) = \frac{1}{N}\sum_{i=1}^{N}(x_i - E(X))(y_i - E(Y))$$

- Here, x, and y, are data values from *X* and Y.
- E(X) and E(Y) are the mean values of x_i and y_i
- *N* is the number of given data.
- Also, the COV(X, Y) is same as COV(Y, X).

Bivariate Data - Covariance

$$\text{cov}(X, Y) = \frac{1}{N}\sum_{i=1}^{N}(x_i - E(X))(y_i - E(Y))$$

Example 2.6: Find the covariance of data $X = \{1, 2, 3, 4, 5\}$ and $Y = \{1, 4, 9, 16, 25\}$.

Solution: Mean(X) = $E(X) = \frac{15}{5} = 3$, Mean(Y) = $E(Y) = \frac{55}{5} = 11$. The covariance is computed using Eq. (2.17) as:

$$\frac{(1-3)(1-11)+(2-3)(4-11)+(3-30)(9-11)+(4-3)(16-11)+(5-3)(25-11)}{5} = 12$$

The covariance between X and Y is 12. It can be normalized to a value between −1 and +1. This is done by dividing it by the correlation of variables. This is called Pearson correlation coefficient. Sometimes, $N - 1$ is also can be used instead of N. In that case, the covariance is $60/4 = 15$.

Bivariate Data-Correlation

- The **Pearson correlation coefficient** is the most common test for **determining any association between two phenomena.**
- It measures the strength and direction of a *linear* relationship between the *x* and y variables.
- The correlation indicates the **relationship between dimensions using its sign.**
- The **sign is more important** than the actual value.
- If the value is positive, it indicates that the dimensions increase together.
- If the value is negative, it indicates that while one-dimension increases, the other dimension decreases.
- If the value is zero, then it indicates that both the dimensions are independent of each other.
- If the dimensions are correlated, then it is better to remove one dimension as it is a redundant dimension.

Bivariate Data-Correlation

If the given attributes are $X = (x_1, x_2, \cdots, x_N)$ and $Y = (y_1, y_2, \cdots, y_N)$, then the Pearson correlation coefficient, that is denoted as r, is given as:

$$r = \frac{COV(X,Y)}{\sigma_X \sigma_Y} \tag{2.18}$$

where, σ_X, σ_Y are the standard deviations of X and Y.

Example 2.7: Find the correlation coefficient of data $X = \{1, 2, 3, 4, 5\}$ and $Y = \{1, 4, 9, 16, 25\}$.

Solution: The mean values of X and Y are $\frac{15}{5} = 3$ and $\frac{55}{5} = 11$. The standard deviations of X and Y are 1.41 and 8.6486, respectively. Therefore, the correlation coefficient is given as ratio of covariance (12 from the previous problem 2.5) and standard deviation of x and y as per Eq. (2.18) as:

$$r = \frac{12}{1.41 \times 8.6486} \approx 0.984$$

MULTIVARIATE STATISTICS

- Multivariate data is the **analysis of more than two observable variables.**
- Some of the multivariate analysis are **regression analysis, principal component analysis, and path analysis.**

Id	Attribute 1	Attribute 2	Attribute 3
1	1	4	1
2	2	5	2
3	3	6	1

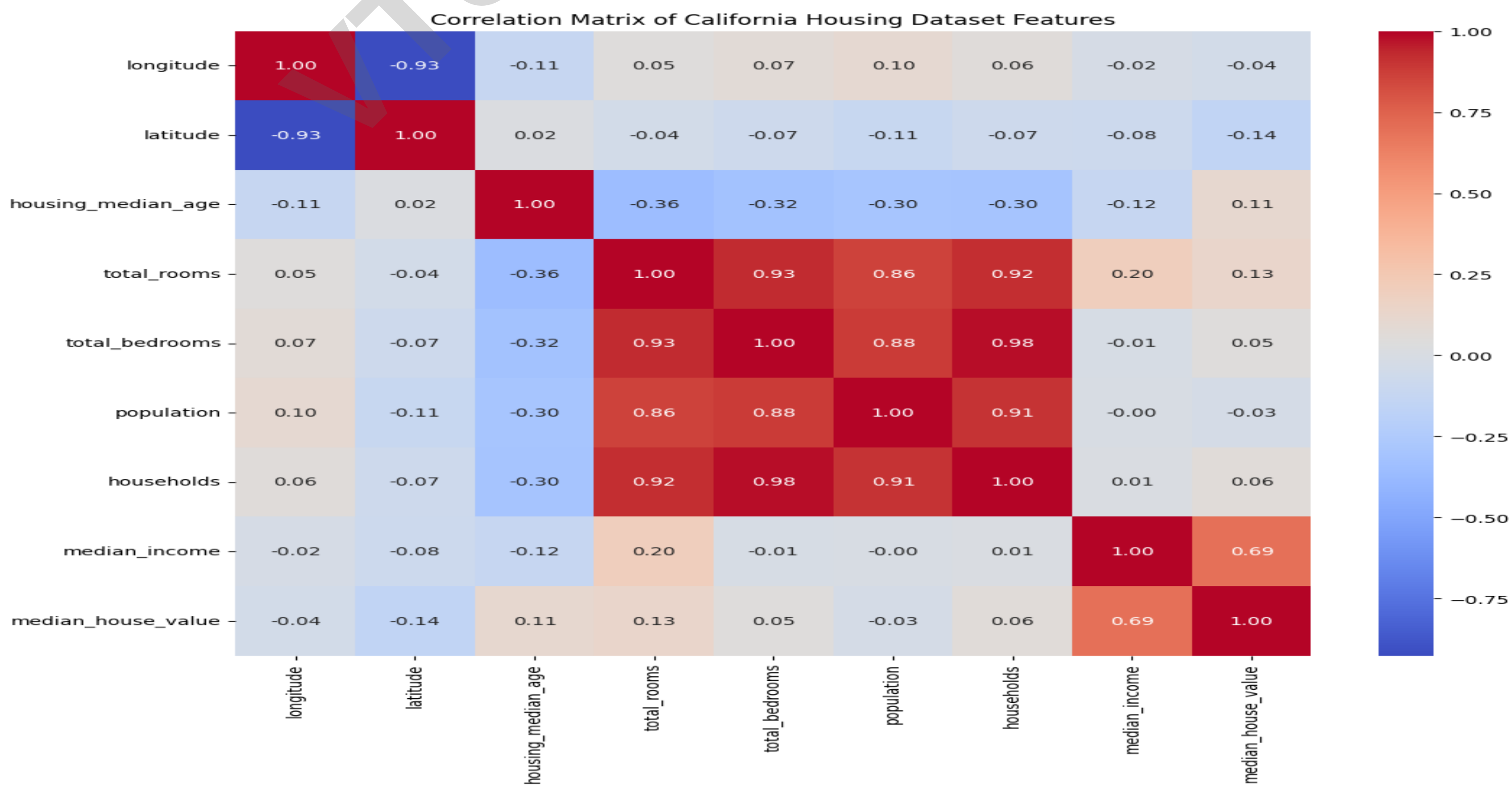
- The mean of multivariate data is a mean vector
- The mean of the above three attributes is given as

  (2, 7.5, 1.33).
- The variance of multivariate data becomes the covariance matrix.
- The mean vector is called centroid and variance is called dispersion matrix.

Multivariate Data Visualization

Heatmap

- Heatmap is a graphical representation of 2D matrix.
- It takes a matrix as input and colours it.
- The darker colours indicate very large values and lighter colours indicate smaller values.



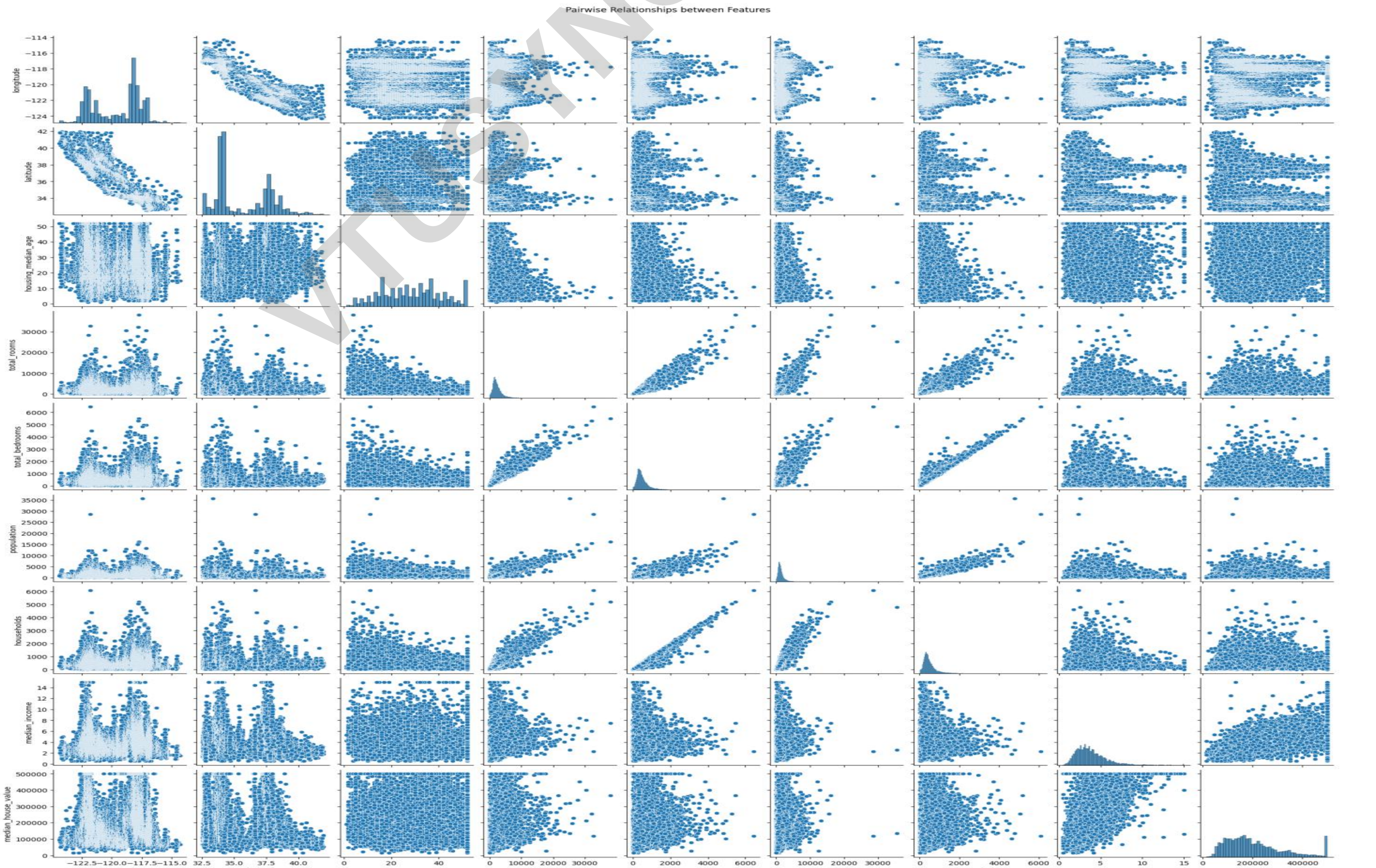
Multivariate Data Visualization

Pairplot

- Pairplot or scatter matrix is a data visual technique for multivariate data.
- A scatter matrix consists of several **pair-wise scatter plots of variables of the multivariate data.**
- All the results are presented in a matrix format.
- By visual examination of the chart, one can easily **find relationships among the variables such as correlation between the variables.**

Multivariate Data Visualization

Pairplot



ESSENTIAL MATHEMATICS FOR MULTIVARIATE DATA

- Machine learning involves many mathematical concepts from the domain of **Linear algebra, Statistics, Probability and Information theory.**
- **'Linear Algebra'** is a branch of mathematics that is central for many **scientific applications**
- Linear algebra deals with **linear equations, vectors, matrices, vector spaces and transformations**.
- These are the **driving forces of machine learning** and machine learning cannot exist without these data types.

Linear Systems and Gaussian Elimination for Multivariate Data

A linear system of equations is a group of equations with unknown variables.

Let $Ax = y$, then the solution x is given as:

$$x = y/A = A^{-1} y \tag{2.19}$$

This is true if y is not zero and A is not zero. The logic can be extended for N-set of equations with 'n' unknown variables.

It means if $A = \begin{pmatrix} a_{11} & a_{12} & \cdots a_{1n} \\ \vdots & \vdots & \vdots & \vdots \\ a_{1n} & a_{2n} & \cdots a_{nn} \end{pmatrix}$ and $y = (y_1 \;\; y_2 \cdots y_n)$, then the unknown variable x can be computed as:

$$x = y/A = A^{-1} y \tag{2.20}$$

If there is a unique solution, then the system is called consistent independent. If there are various solutions, then the system is called consistent dependant. If there are no solutions and if the equations are contradictory, then the system is called inconsistent.

Gaussian Elimination for Multivariate Data

- For solving large number of system of equations, Gaussian elimination can be used.
- **The procedure for applying Gaussian elimination is given as follows:**

1. Write the given matrix.
2. Append vector y to the matrix A. This matrix is called augmentation matrix.
3. Keep the element a_{11} as pivot and eliminate all a_{11} in second row using the matrix operation, $R_2 - \left(\frac{a_{21}}{a_{11}}\right)$, here R_2 is the second row and $\left(\frac{a_{21}}{a_{11}}\right)$ is called the multiplier. The same logic can be used to remove a_{11} in all other equations.
4. Repeat the same logic and reduce it to reduced echelon form. Then, the unknown variable as:
$$x_n = \frac{y_{nn}}{a_{nn}} \tag{2.21}$$
5. Then, the remaining unknown variables can be found by back-substitution as:
$$x_{n-1} = \frac{y_{n-1} - a_{n-1} \times x_n}{a_{(n-1)(n-1)}} \tag{2.22}$$
This part is called backward substitution.

- To facilitate the application of Gaussian elimination method, the following row operations are applied:
- Swapping the rows
- Multiplying or dividing a row by a constant
- Replacing a row by adding or subtracting a multiple of another row to it

Example 2.8: Solve the following set of equations using Gaussian Elimination method.

$$2x_1 + 4x_2 = 6$$
$$4x_1 + 3x_2 = 7$$

Solution: Rewrite this in matrix form as follows:

$$\left(\begin{array}{cc|c} 2 & 4 & 6 \\ 4 & 3 & 7 \end{array}\right)$$

$$\sim \left(\begin{array}{cc|c} 2 & 4 & 6 \\ 4 & 3 & 7 \end{array}\right) R_1 = \frac{R_1}{2}$$

Apply the transformation by dividing the row 1 by 2. There are no general guidelines of row operations other than reducing the given matrix to row echelon form. The operator ~ means reducing to. The above matrix can further be reduced as follows:

$$\sim \left(\begin{array}{cc|c} 1 & 2 & 3 \\ 4 & 3 & 7 \end{array}\right) R_2 = R_2 - 4R_1$$

$$\sim \left(\begin{array}{cc|c} 1 & 2 & 3 \\ 0 & -5 & -5 \end{array}\right) R_2 = R_2 / -5$$

$$\sim \left(\begin{array}{cc|c} 1 & 2 & 3 \\ 0 & 1 & 1 \end{array}\right) R_1 = R_1 - 2R_2$$

$$\sim \left(\begin{array}{cc|c} 1 & 0 & 1 \\ 0 & 1 & 1 \end{array}\right)$$

Therefore, in the reduced echelon form, it can be observed that:

$$x_2 = 1$$
$$x_1 = 1$$

Matrix Decompositions

- It is often necessary to reduce a matrix to its constituent parts so that complex matrix operations can be performed. These methods are also known as matrix factorization methods.
- The most popular matrix decomposition is called **eigen decomposition.**
- It is a way of **reducing the matrix into eigen values and eigen vectors.**
- Then, the matrix *A* can be decomposed as:

$$A = Q \Lambda Q^T$$

- where, Q is the matrix of eigen vectors, A is the diagonal matrix and Q^T is the transpose of matrix *Q.*

LU Decomposition

- One of the simplest matrix decompositions is *LU* decomposition where the matrix A can be decomposed matrices: ***A = LU***
- Here, L is the lower triangular matrix and *U* is the upper triangular matrix.
- The decomposition can be done using Gaussian elimination method.
- First, an identity matrix is augmented to the given matrix.
- Then, row operations and Gaussian elimination is applied to reduce the given matrix to get matrices *L* and *U.*

Example 2.9: Find LU decomposition of the given matrix:

$$A = \begin{pmatrix} 1 & 2 & 4 \\ 3 & 3 & 2 \\ 3 & 4 & 2 \end{pmatrix}$$

Solution: First, augment an identity matrix and apply Gaussian elimination. The steps are as shown in:

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 2 & 4 \\ 3 & 3 & 2 \\ 3 & 4 & 2 \end{bmatrix} \qquad \boxed{\text{Initial Matrix}}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 2 & 4 \\ 0 & -3 & -10 \\ 3 & 4 & 2 \end{bmatrix} \qquad \boxed{R_2 = R_2 - 3R_1}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 3 & 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 2 & 4 \\ 0 & -3 & -10 \\ 0 & -2 & -10 \end{bmatrix} \qquad \boxed{R_3 = R_3 - 3R_1}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 3 & \frac{2}{3} & 1 \end{bmatrix}\begin{bmatrix} 1 & 2 & 4 \\ 0 & -3 & -10 \\ 0 & 0 & \frac{-10}{3} \end{bmatrix} \qquad \boxed{R_3 = R_3 - \frac{2}{3}R_2}$$

Now, it can be observed that the first matrix is L as it is the lower triangular matrix whose values are the determiners used in the reduction of equations above such as 3, 3 and 2/3. The second matrix is U, the upper triangular matrix whose values are the values of the reduced matrix because of Gaussian elimination.

$$L = \begin{pmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 3 & \frac{2}{3} & 1 \end{pmatrix} \text{ and } U = \begin{pmatrix} 1 & 2 & 4 \\ 0 & -3 & -10 \\ 0 & 0 & -\frac{10}{3} \end{pmatrix}.$$

Machine Learning and Importance of Probability and Statistics

- Like linear algebra, **statistics is the heart of machine learning.**
- The importance of statistics needs to be stressed as **without statistics analysis of data is difficult.**
- Probability is especially important for machine learning.
- **Any data can be assumed to be generated by a probability distribution.**
- So, **a knowledge of probability distribution and random variables are must** for better understanding of the machine learning concepts.

Machine Learning and Importance of Probability and Statistics

- This is about **hypothesis or constructing a model.**
- So, **the theory of hypothesis, construction of models, evaluating the models using hypothesis testing,** significance of the model and evaluation of the model are all linking factors of machine learning and probability theory.
- Similarly, the **construction of datasets using sampling theory** is also required.

Probability Distributions

- A probability distribution of a variable, say X, summarizes the probability associated with X's events.
- In other words, **distribution is a function that describes the relationship between the observations in a sample space.**
- Probability distributions are of two types:
  - Discrete probability distribution
  - Continuous probability distribution
- **The relationships between the events for a continuous random variable and their probabilities is called a continuous probability distribution.**
- It is summarized as **Probability Density Function (PDF).**
- PDF calculates the probability of observing an instance.
- The plot of PDF shows the shape of the distribution.
- **Cumulative Distributive Function (CDF)** computes the probability of an observation value.
- **Continuous Probability Distributions: Normal, Rectangular, and Exponential distributions fall under this category.**

Normal Distribution

- Normal distribution is a **continuous probability distribution.**
- **This is also known as gaussian distribution or bell-shaped curve distribution.**
- It is the most common distribution function.
- The **shape** of this distribution is a typical **bell-shaped curve.**
- In normal distribution, **data tends to be around a central value with no bias on left or right.**

PDF of the normal distribution is given as:

$$f(x, \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \tag{2.24}$$

Here, μ is mean and σ is the standard deviation. Normal distribution is characterized by two parameters – mean and variance.

Normal Distribution

- One important concept associated with normal distribution is z-score.
- It can be computed as:

$z = \frac{x - \mu}{\sigma}$. When μ is zero and σ is 1, z-score is same as x. This is useful to normalize the data.

Rectangular Distribution

- This is also known as **uniform distribution.**
- It has equal probabilities for all values in the range *a, b.*
- The uniform distribution is given as follows:

$$P(X = x) = \begin{cases} \dfrac{1}{b-a} & \text{for } a \le x \le b \\ 0 & \text{Otherwise} \end{cases}$$

Exponential Distribution

- This is a **continuous uniform distribution.**
- This probability distribution is used to **describe the time between events in a Poisson process.**
- Exponential distribution is another special case of **Gamma distribution with a fixed parameter of 1.**

The PDF is given as follows:

$$f(x, \lambda) = \begin{cases} \lambda e^{-\lambda x} & \text{if } x \geq 0 \quad (\lambda > 0) \\ 0 & \text{if } x < 0 \end{cases} \tag{2.26}$$

Here, x is a random variable and λ is called rate parameter. The mean and standard deviation of exponential distribution is given as β, where, $\beta = \frac{1}{\lambda}$.

Discrete Distribution

- Binomial, Poisson, and Bernoulli distributions fall under this category.
- **1. Binomial Distribution -** Binomial distribution is another distribution that is often encountered in machine learning
- **It has only two outcomes: success or failure.**
- It is also called **Bernoulli trial.**
- The **objective** of this distribution is to **find probability of getting success *k* out of n trials.**
- The way to get success out of *k* out of n number of trials is given as:

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

The binomial distribution function is given as follows, where p is the probability of success and probability of failure is $(1 - p)$. The probability of success in a certain number of trials is given as:

$$p^k(1 - p)^{n-k} \text{ or } p^k q^{n-k} \tag{2.28}$$

Combining both, one gets PDF of binomial distribution as:

$$\binom{n}{k} p^k (1 - p)^{n-k} \tag{2.29}$$

2. Poisson Distribution

- Given an interval of time, this distribution is used **to model the probability of a given number of events *k*.**

The PDF of Poisson distribution is given as follows:

$$f(X = x; \lambda) = Pr[X = x] = \frac{e^{-\lambda}\lambda^{x}}{x!} \qquad (2.33)$$

Here, x is the number of times the event occurs and λ is the mean number of times an event occurs.

The mean is the population mean at number of emails received and the standard deviation is $\sqrt{\lambda}$.

3. Bernoulli Distribution

- This distribution models an experiment whose **outcome is binary.**
- The outcome is positive with *p* and negative with 1 - *p.*

The outcome is positive with p and negative with $1 - p$. The PMF of this distribution is given as:

$$f(k;p) = \begin{cases} q = 1 - p & \text{if } k = 0 \\ p & \text{if } k = 1. \end{cases} \tag{2.34}$$

The mean is p and variance is $p(1 - p) = q$

FEATURE ENGINEERING AND DIMENSIONALITY REDUCTION TECHNIQUES

- Features are attributes.
- **Feature engineering is about determining the subset of features that form an important part of the input that improves the performance of the model.**
- Feature engineering deals with two problems — **Feature Transformation and Feature Selection.**
- **Feature transformation is extraction of features and creating new features that may be helpful in increasing performance.**
- For example, the height and weight may give a new attribute called Body Mass Index (BMI).
- **Feature subset selection** is another important aspect of feature engineering that **focuses on selection of features to reduce the time but not at the cost of reliability.**

FEATURE ENGINEERING AND DIMENSIONALITY REDUCTION TECHNIQUES

- **The subset selection reduces the dataset size by removing irrelevant features** and constructs a minimum set of attributes for machine learning.
- If the dataset has n attributes, then time complexity is extremely high as n dimensions need to be processed for the given dataset.
- For n attributes, there are 2" possible subsets.
- If the value of n is high, the problem becomes inflexible. This is called **'curse of dimensionality'.**
- Since, **as the number of dimensions increases, the time complexity increases.** The remedy is that some of the components that do not contribute much can *be* deleted. This results in the **reduction of dimensionality.**
- **Choosing optimal attributes becomes a graph search problem.**

Characteristics of Good Features

- The features can be removed based on two aspects:
- **1. Feature relevancy -** Some features contribute more for classification than other features.
- For example, a mole on the face can help in face detection than common features like nose.
- In simple words, the features should be relevant.
- **2. Feature redundancy -** Some features are redundant.
- For example, when a database table has a field called Date of birth, then age field is not relevant as age can be computed easily from date of birth.
- This helps in removing the column age that leads to reduction of dimension one.
- So, the procedure is:
  - Generate all possible subsets
  - Evaluate the subsets and model performance
  - Evaluate the results for optimal feature selection

Feature Selection Algorithms

- **Stepwise Forward Selection:**
  - This procedure starts with an empty set of attributes.
  - Every time, an attribute is **tested for statistical significance for best quality and is added to the reduced set.**
  - This process is continued till a good reduced set of attributes is obtained.

Feature Selection Algorithms

- **Stepwise Backward Elimination**
  - This procedure starts with a complete set of attributes.
  - At every stage, the procedure **removes the worst attribute from the set, leading to the reduced set.**
  - **Combined Approach** Both forward and reverse methods can be combined so that the procedure can add the best attribute and remove the worst attribute.

Feature Selection Algorithms:
Principal Component Analysis(PCA)

- **What is PCA?**
  - A statistical technique for dimensionality reduction.
  - Transforms high-dimensional data into a lower-dimensional space.
- **Why PCA?**
  - Reduces complexity.
  - Removes redundancy in data.
  - Improves visualization and computational efficiency.

Key Concepts

- **Variance**: PCA maximizes variance in the data.
- **Covariance Matrix**: Measures relationships between features.
- **Eigenvalues and Eigenvectors**: Used to identify principal components.
- **Principal Components**: Orthogonal directions capturing maximum variance.

Steps in PCA

- **Standardize the Data**: Scale features to have zero mean and unit variance.
- **Compute Covariance Matrix**: Understand relationships between features.
- **Calculate Eigenvalues and Eigenvectors**: Identify principal components.
- **Sort and Select Components**: Choose top k components based on explained variance.
- **Transform Data**: Project data onto the new subspace.

PCA

Consider a group of random vectors of the form:

$$x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

The *mean vector* of the set of random vectors is defined as:

$$m_x = E\{x\}$$

- The operator *E* refers to the expected value of the population.
- This is calculated theoretically using the probability density functions (PDF) of the elements x_i & x_j.

PCA

Compute Covariance matrix a $C = E\{(x - m_x)(x - m_x)^T\}$

Compute Eigen values and Eigen vectors and matrix A as a set of eigen vectors

For M random vectors, when M is large enough, the mean vector and covariance matrix can be approximately calculated as:

$$m_x = \frac{1}{M}\sum_{k=1}^{M} x_k \tag{2.53}$$

$$A = \frac{1}{M}\sum_{k=1}^{M} x_k x_k^T - m_x m_x^T \tag{2.54}$$

PCA

Compute PCA as

The mapping of the vectors x to y using the transformation can now be described as:

$$y = A(x - m_x) \tag{2.55}$$

This transform is also called as Karhunen-Loeve or Hoteling transform. The original vector x can now be reconstructed as follows:

$$x = A^T y + m_x \tag{2.56}$$

Applications of PCA

- **Data Compression**: Reduce storage and computation costs.
- **Visualization**: Project high-dimensional data into 2D or 3D.
- **Noise Reduction**: Remove less important components.
- **Feature Engineering**: Create new features for machine learning models.

PCA Algorithm

1. The target dataset x is obtained
2. The mean is subtracted from the dataset. Let the mean be m. Thus, the adjusted dataset is $X - m$. The objective of this process is to transform the dataset with zero mean.
3. The covariance of dataset x is obtained. Let it be C.
4. Eigen values and eigen vectors of the covariance matrix are calculated.
5. The eigen vector of the highest eigen value is the principal component of the dataset. The eigen values are arranged in a descending order. The feature vector is formed with these eigen vectors in its columns.

   Feature vector = {eigen vector$_1$, eigen vector$_2$, … , eigen vector$_n$}
6. Obtain the transpose of feature vector. Let it be A.
7. PCA transform is $y = A \times (x - m)$, where x is the input dataset, m is the mean, and A is the transpose of the feature vector.

The original data can be retrieved using the formula given below:

$$\text{Original data } (f) = \{(A)^{-1} \times y\} + m \tag{2.58}$$

$$= \{(A)^{T} \times y\} + m \tag{2.59}$$

Example 2.12: Let the data points be $\begin{pmatrix} 2 \\ 6 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ 7 \end{pmatrix}$. Apply PCA and find the transformed data. Again, apply the inverse and prove that PCA works.

Solution: One can combine two vectors into a matrix as follows:

The mean vector can be computed as Eq. (2.53) as follows:

$$\mu = \begin{pmatrix} \frac{2+1}{2} \\ \frac{6+7}{2} \end{pmatrix} = \begin{pmatrix} 1.5 \\ 6.5 \end{pmatrix}$$

As part of PCA, the mean must be subtracted from the data to get the adjusted data:

$$x_1 = \begin{pmatrix} 2-1.5 \\ 6-6.5 \end{pmatrix} = \begin{pmatrix} 0.5 \\ -0.5 \end{pmatrix}$$

$$x_2 = \begin{pmatrix} 1-1.5 \\ 7-6.5 \end{pmatrix} = \begin{pmatrix} -0.5 \\ 0.5 \end{pmatrix}$$

One can find the covariance for these data vectors. The covariance can be obtained using Eq. (2.54):

$$m_1 = \begin{pmatrix} 0.5 \\ -0.5 \end{pmatrix} \begin{pmatrix} 0.5 & -0.5 \end{pmatrix} = \begin{pmatrix} 0.25 & -0.25 \\ -0.25 & 0.25 \end{pmatrix}$$

$$m_2 = \begin{pmatrix} -0.5 \\ 0.5 \end{pmatrix} \begin{pmatrix} -0.5 & 0.5 \end{pmatrix} = \begin{pmatrix} 0.25 & -0.25 \\ -0.25 & 0.25 \end{pmatrix}$$

The final covariance matrix is obtained by adding these two matrices as:

$$C = \begin{pmatrix} 0.5 & -0.5 \\ -0.5 & 0.5 \end{pmatrix}$$

The eigen values and eigen vectors of matrix C can be obtained as $\lambda_1 = 1$, $\lambda_1 = 0$. The eigen vectors are $\begin{pmatrix} -1 \\ 1 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$.

$\lambda_1 = 0$. The eigen vectors are $\begin{pmatrix} -1 \\ 1 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$. The matrix A can be obtained by packing the eigen vector of these eigen values (after sorting it) of matrix C. For this problem, $A = \begin{pmatrix} -1 & 1 \\ 1 & 1 \end{pmatrix}$. The transpose of A, $A^T = \begin{pmatrix} -1 & 1 \\ 1 & 1 \end{pmatrix}$ is also the same matrix as it is an orthogonal matrix. The matrix can be normalized by diving each elements of the vector, by the norm of the vector to get:

$$A = \begin{pmatrix} -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix}$$

One can check that the PCA matrix A is orthogonal. A matrix is orthogonal is $A^{-1} = A$ and $AA^{-1} = I$.

$$AA^T = \begin{pmatrix} -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

The transformed matrix y using Eq. (2.55) is given as:

$$y = A \times (x - m)$$

Recollect that $(x-m)$ is the adjusted matrix.

$$y = A(x-m) = \begin{pmatrix} -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} 0.5 & -0.5 \\ -0.5 & 0.5 \end{pmatrix}$$

$$= \begin{pmatrix} -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} \end{pmatrix} \left(\textit{for convenience } 0.5 = \frac{1}{2} \right)$$

$$= \begin{pmatrix} -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ 0 & 0 \end{pmatrix}$$

One can check the original matrix can be retrieved from this matrix as:

$$\{(A)^T \times y\} + m$$

$$x = A^T y + m = \begin{pmatrix} -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ 0 & 0 \end{pmatrix} + \begin{pmatrix} 1.5 \\ 6.5 \end{pmatrix}$$

$$= \begin{pmatrix} \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} \end{pmatrix} + \begin{pmatrix} 1.5 \\ 6.5 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 6 & 7 \end{pmatrix}$$

Therefore, one can infer the original is obtained without any loss of information.

Step 1: Compute the Characteristic Equation

The characteristic equation is given by:

$$\det(A - \lambda I) = 0$$

$$\begin{vmatrix} 0.5 - \lambda & -0.5 \\ -0.5 & 0.5 - \lambda \end{vmatrix} = 0$$

Expanding the determinant:

$$(0.5 - \lambda)(0.5 - \lambda) - (-0.5)(-0.5) = 0$$

$$(0.5 - \lambda)^2 - 0.25 = 0$$

$$0.25 - \lambda + \lambda^2 - 0.25 = 0$$

$$\lambda^2 - \lambda = 0$$

$$\lambda(\lambda - 1) = 0$$

So, the eigenvalues are:

$$\lambda_1 = 0, \quad \lambda_2 = 1$$

Step 2: Compute the Eigenvectors

For each eigenvalue λ, we solve $(A - \lambda I)x = 0$.

For $\lambda_1 = 0$:

$$\begin{bmatrix} 0.5 & -0.5 \\ -0.5 & 0.5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$$

$$\begin{bmatrix} 0.5 & -0.5 \\ -0.5 & 0.5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

This simplifies to:

$$0.5x_1 - 0.5x_2 = 0$$

$$x_1 = x_2$$

Thus, the eigenvector corresponding to $\lambda_1 = 0$ is:

$$\begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

For $\lambda_2 = 1$:

$$(A - I)x = 0$$

$$\begin{bmatrix} -0.5 & -0.5 \\ -0.5 & -0.5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

This simplifies to:

$$-0.5x_1 - 0.5x_2 = 0$$

$$x_1 = -x_2$$

Thus, the eigenvector corresponding to $\lambda_2 = 1$ is:

$$\begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Final Answer

- Eigenvalues: $\lambda_1 = 0, \lambda_2 = 1$
- Eigenvectors:
  - For $\lambda_1 = 0$: $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$
  - For $\lambda_2 = 1$: $\begin{bmatrix} 1 \\ -1 \end{bmatrix}$

Linear Discriminant Analysis (LDA)

- Linear Discriminant Analysis (LDA) is also a **feature reduction technique like PCA.**
- The focus of LDA is to **project higher dimension data to a line (lower dimension data).**
- LDA is also used to classify the data.

Linear Discriminant Analysis Linear Discriminant Analysis (LDA)

- Let there be two classes, c1 and c2.
- Let μ_1 & μ_2 be the mean of the patterns of two classes.
- The mean of the class c1 and c2 can be computed as:

$$\mu_1 = \frac{1}{N_1} \sum_{x_i \in c_1}^{n} x_i \text{ and } \mu_2 = \frac{1}{N_2} \sum_{x_i \in c_2}^{n} x_i$$

The aim of LDA is to optimize the function:

$$J(V) = \frac{V^T \sigma_B V}{V^T \sigma_W V} \tag{2.60}$$

where, V is the linear projection and σ_B and σ_W are class scatter matrix and within scatter matrix, respectively. For the two-class problem, these matrices are given as:

$$\sigma_B = N_1(\mu_1 - \mu)(\mu_1 - \mu)^T + N_2(\mu_2 - \mu)(\mu_2 - \mu)^T \tag{2.61}$$

$$\sigma_W = \sum_{x_i \in c_1} (x_i - \mu_1)(x_i - \mu_1)^T + \sum_{x_i \in c_2} (x_i - \mu_2)(x_i - \mu_2)^T \tag{2.62}$$

The maximization of $J(V)$ should satisfy the equation:

$$\sigma_B V = \lambda \sigma_W V \text{ or } \sigma_W^{-1} \sigma_B V = \lambda V \tag{2.63}$$

As $\sigma_B V$ is always in the direction of $(\mu_1 - \mu_2)$, V can be given as:

$$V = \sigma_W^{-1}(\mu_1 - \mu_2) \tag{2.64}$$

Let $V = \{v_1, v_2, \cdots, v_d\}$ be the generalized eigen vectors of σ_B and σ_W, where, d is the largest eigen values as in PCA. The transformation of x is then given as:

$$y = V_d^T x \tag{2.65}$$

Like in PCA, the largest eigen values can be retained to have projections.

Singular Value Decomposition (SVD)

- Singular Value Decomposition (SVD) is another useful decomposition technique.
- Let A be the matrix, then the matrix A can be decomposed as:

$$A = USV^T$$

- Here, A is the given matrix of dimension m x n,
- U is the orthogonal matrix whose dimension is

  m x n,
- S is the diagonal matrix of dimension mx n, and V is the orthogonal matrix.

Singular Value Decomposition Singular Value Decomposition (SVD)

- The procedure for finding decomposition matrix is given as follows:
  1. For a given matrix, find AA^T
  2. Find eigen values of AA^T
  3. Sort the eigen values in a descending order. Pack the eigen vectors as a matrix U.
  4. Arrange the square root of the eigen values in diagonal. This matrix is diagonal matrix, S.
  5. Find eigen values and eigen vectors for A^TA. Find the eigen value and pack the eigen vector as a matrix called V.
- Thus, $A = USV^T$.
- Here, U and V are orthogonal matrices. The columns of U and V are left and right singular values, respectively.
- SVD is useful in compression, as one can decide **to retain only a certain component instead of the original matrix A as:**

$$a_{ij} = \sum_{k=1}^{n} u_{ik} s_k v_{jk}$$

SVD Example

Example 2.13: Find SVD of the matrix:

$$A = \begin{pmatrix} 1 & 2 \\ 4 & 9 \end{pmatrix}$$

Solution: The first step is to compute:

$$AA^T = \begin{pmatrix} 1 & 2 \\ 4 & 9 \end{pmatrix}\begin{pmatrix} 1 & 4 \\ 2 & 9 \end{pmatrix} = \begin{pmatrix} 5 & 22 \\ 22 & 97 \end{pmatrix}$$

The eigen value and eigen vector of this matrix can be calculated to get U. The eigen values of this matrix are 0.0098 and 101.9902.

The eigen vectors of this matrix are:

$$u_1 = \begin{pmatrix} 0.2268 \\ 1 \end{pmatrix}$$

$$u_2 = \begin{pmatrix} -4.4086 \\ 1 \end{pmatrix}$$

These vectors are normalized to get the vectors respectively as:

$$u_1 = \begin{pmatrix} 0.2212 \\ 0.9752 \end{pmatrix}$$

$$u_2 = \begin{pmatrix} -0.9752 \\ 0.2212 \end{pmatrix}$$

The matrix U can be obtained by concatenating the above vector as:

$$U = [u_1, u_2] = \begin{pmatrix} 0.2212 & -0.9752 \\ 0.9752 & 0.2212 \end{pmatrix}$$

The matrix V can be obtained by finding A^TA. It is $\begin{pmatrix} 17 & 38 \\ 38 & 85 \end{pmatrix}$. The eigen values are 0.0098 and 101.9902. The eigen vectors can be found as follows:

$$v_1 = \begin{pmatrix} 0.447 \\ 1 \end{pmatrix} \text{ when } \lambda = 101.99$$

$$v_2 = \begin{pmatrix} -2.236 \\ 1 \end{pmatrix} \text{ when } \lambda = 0.0098$$

The above can be normalized as follows:

$$v_1 = \begin{pmatrix} 0.4082 \\ 0.9129 \end{pmatrix}$$

$$v_2 = \begin{pmatrix} -0.9129 \\ 0.4082 \end{pmatrix}$$

The matrix V can be obtained by concatenating the above vector as:

$$V = [v_1 \;\; v_2] = \begin{pmatrix} 0.4081 & -0.9129 \\ 0.9129 & 0.4082 \end{pmatrix}$$

The matrix S can be found as the diagonal matrix as:

$$S = \begin{pmatrix} \sqrt{101.9902} & 0 \\ 0 & \sqrt{0.0098} \end{pmatrix} = \begin{pmatrix} 10.099 & 0 \\ 0 & 0.099 \end{pmatrix}$$

Therefore, the matrix decomposition $A = U\,SV^T$ is complete.